Predicting Mental Health Trends

G11-NeuroNexus

Data Science Capstone Project
Launch Report

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Team Members:

Name: Sanjoli Sogani

Name: Aman Ostwal

Name: Sai Pokuri

Name: Darshit Rai

The System/Product

System/Product Name: Predicting Mental Health Trends

Introduction:

In recent years, mental health has gained significant global attention. Platforms like Reddit have become essential forums where individuals openly share their experiences, discuss challenges, and seek advice on mental health issues. These platforms house vast amounts of unstructured data, offering a unique opportunity to uncover insights into mental health trends, triggers, and coping mechanisms. By analyzing these discussions, we can better understand public sentiment and identify emerging patterns in mental health.

Deliverables:

- **Data Acquisition:** Collection and preprocessing of Reddit data, including posts, comments, and engagement metrics from mental health-related subreddits.
- **Data Preprocessing:** Cleaning data by removing duplicates and noisy data. Text preprocessing which includes tokenization and lemmatization.
- **EDA:** Detailed analysis of trends, examination of emotional tones in mental health discussions – visualizing them through graphs.
- **Final Report:** Documenting the data acquisition process and analysis which could be used for contributing increased awareness.

Highlighted Features:

Gather Data Using Reddit API -

- Use the Reddit API (via the PRAW library in Python, Google Colab) to collect data from mental health-focused subreddits, such as:
  - r/anxiety
  - r/depression
  - r/adhd
- Extract post content, comments, timestamps, and other metadata for analysis.

Clean and Filter Data -

- Remove Irrelevant Data: Eliminate posts or comments that do not contain meaningful text (e.g., URLs, advertisements, or automated messages).
- Handle Missing Data: Remove entries with missing or incomplete text fields.
- Standardize Text:
  - Convert all text to lowercase.
  - Remove special characters, numbers, and punctuation.
  - Expand contractions (e.g., "can't" to "cannot").

- Remove Stop Words: Filter out common stop words (e.g., *and, the, is*) using an NLP library like NLTK or SpaCy.
- Lemmatization/Stemming: Reduce words to their root forms to standardize variations (e.g., *struggled* → *struggle*).

Create an Algorithm for Severity Scoring -

- We are going to develop an algorithm to define severity levels for mental health discussions. This algorithm will be informed by keywords commonly associated with mental health, with severity levels determined based on insights from research papers and reputable sources. The algorithm will dynamically assign weights to these keywords, classifying them into high, medium, and low severity categories, ensuring the approach is evidence-based and reflective of current research.
  - High Severity
  - Medium Severity
  - Low Severity

Analyze Sentiment Over Time -

- Track sentiment fluctuations across timeframes to identify trends in community well-being. Highlight periods of increased negative sentiment or high severity scores might correlate with external factors (e.g., world events, holidays).
- Create dashboards or visual reports showing trends, sentiment shifts, and key discussion topics.
- Use machine learning to predict potential mental health trends based on past data.

Hybrid Learning Approach -

To enhance the accuracy and flexibility of the analysis, we will adopt a hybrid learning approach that combines supervised and unsupervised methods.

- Supervised Learning Topics:

  - Severity Classification: Train models to classify mental health discussions into predefined severity levels (e.g., high, medium, low).
  - Topic-Specific Classification: Predict the primary mental health issue discussed in a post (e.g., anxiety, depression, ADHD).
  - We will use Random Forest for robust, non-linear classification with feature importance for interpreting severity indicators and XGBoost for high accuracy in handling imbalanced datasets and boosting performance. For hyperparameter tuning, we will employ grid search for systematic exploration or random search for efficient sampling within the parameter space.

- Unsupervised Learning Topics:

  - We will use techniques like Latent Dirichlet Allocation (LDA) to uncover latent themes in mental health discussions and clustering for grouping similar posts based on their textual content or sentiment patterns.

Target Variable -

We will define two primary target variables:

- The type of mental health disorder a post indicates.
- The severity level of the mental health condition based on post content.

Sponsor or Proxy User:

None for now

Issues:

None for now

Potential Issues in Data Acquisition for Predicting Mental Health Trends:

Access to Data -

- Reddit has strict guidelines; data scraping rate limits and may require multiple authentication restrictions.
- Some subreddits might have rules that restrict API access, we may need approval from the moderators to access the data.

Data Quality -

- Reddit posts often contain unstructured data i.e., they may include typos, slang, or abbreviations, which may make it harder to clean and analyze the data.
- Some posts may be irrelevant to the mental health topic or spam, which can add noise and may affect the accuracy of the analysis.

Limited Metadata -

- Reddit data usually comes with basic details like timestamps, engagement etc., it does not provide demographic information like age, gender or location which is essential for gaining deeper insights into the mental health trends.
- Sometimes the user's self-reported data is also incomplete or inaccurate, which makes it difficult to give precise results.

Experts and Stakeholder Requirements:

Professors and Faculty -

- Professors in public health departments can help us in interpreting mental health trends and provide some resources for research. Data science professors can help us with machine learning models and NLP techniques to analyze Reddit data effectively.

University Counselling Centre -

- They can help in sharing anonymized information about common students' mental health conditions and suggest to us the variables to focus on.

Reddit Community Moderators -

- Moderators can grant us permission to access subreddit-specific data that will help us refine the scope of the analysis.

The Team

Team Name: NeuroNexus

Team Members and their Specialties:

Sanjoli Sogani (Team Lead) -

- Will coordinate with all team members to ensure smooth collaboration and workflow.
- Will lead efforts in data pre-processing and cleaning to prepare the dataset for analysis and modeling.

Aman Ostwal -

- Will be responsible to data acquisition by sourcing and preparing the required datasets.
- Will conduct Exploratory Data Analysis (EDA) to uncover insights and trends within the data.

Darshit Rai -

- Will actively contribute to data acquisition by sourcing and preparing the required datasets.
- Will perform Exploratory Data Analysis (EDA) to identify patterns and inform further modeling processes.

Sai Pokuri -

- Will play a pivotal role in data pre-processing, cleaning to ensure the dataset's readiness for analysis.
- Will lead Exploratory Data Analysis (EDA) to extract meaningful insights and trends.
- Will take charge of the modeling phase, designing & implementing predictive and analytical models.

Team Communication:

Weekly Coordination on Teams -

- At the beginning of each week, we communicate on Microsoft Teams to decide the meeting schedule based on everyone's availability.

Scheduled Meetings on Microsoft Teams -

- On the scheduled day, we meet on Microsoft Teams to discuss tasks and deliverables.
- These meetings are brief but focused, ensuring that everyone understands their responsibilities and deadlines.

Mid-Week Follow-Up Meetings -

- After the initial discussion, we plan another meeting later in the week on Teams to review progress and address any challenges.
- This ensures that tasks stay on track and any issues are resolved promptly.

File Sharing and Task Division on Teams -

- All project-related documents and files are shared on Microsoft Teams.
- Tasks are divided among team members during the meeting, and everyone works on their assigned responsibilities.

Peer Review and Quality Check -

- Once tasks are completed, team members review each other's work to ensure quality and consistency.
- This collaborative effort helps maintain ambitious standards and ensures that everyone is aligned with the project goals.

Team Issues:

Data Acquisition Challenges - Difficulty in collecting relevant Reddit data due to API limitations or restrictions.

Approaches

- Allocate time for troubleshooting API access issues and ensure familiarity with tools like PRAW or Pushlift API.

Appendix

Week	Plan	Result
Jan 06, 2025 – Jan 12, 2025	Project Selection	
Jan 13, 2025 – Jan 19, 2025	Finding Data Sources/data and setting up the access to Reddit API for data collection	API setup complete and sample data fetched
Jan 20, 2025 – Jan 26, 2025	Data collection from selected subreddits (e.g. - anxiety, depression) Store data systematically for easy access and processing	Dataset Collection Save in .csv format
Jan 27, 2025 – Feb 02, 2025	Perform data cleaning (e.g. – removal of duplicates, null values, or irrelevant data) Create a preprocessed data Set up meeting with Counselling Centre Advisor to discuss which variables to focus on	Clean dataset ready for analysis
Feb 03, 2025 – Feb 09, 2025	Begin text preprocessing (e.g., tokenization, removal of stop-words, stemming and lemmatization)	Preprocess the dataset
Feb 10, 2025 – Feb 16, 2025	Work on presentation and data acquisition and pre-processing report	Presentation and report completed
Feb 17, 2025 – Feb 23, 2025	Conduct EDA on comments counts, votes Visualization of basic trends	Perform initial EDA
Feb 24, 2025 – Mar 01, 2025	Perform EDA on textual data i.e., commonly used words, phrases, and sentiments Work on word frequencies, patterns, sentiment polarity	Prepare Sentiment trend visualization

Mar 02, 2025 – Mar 08, 2025	Tasks data severity level of each mental health problem Create Exploratory Data Analytics report	Complete EDA EDA report
Mar 09, 2025 – Mar 15, 2025	Refine the project based of feedback Create Status Presentation	Status Presentation

Table of Contributions

The table below identifies contributors to various sections of this document.

	Section	Writing	Editing
1	**Project**	**Everyone**	**Everyone**
2	**Team**	**Everyone**	**Everyone**
3	**Plan**	**Everyone**	**Everyone**

Grading

The grade is given based on quality, clarity, presentation, completeness, and writing of each section in the report. This is the grade of the group. Individual grades will be assigned at the end of the term when peer reviews are collected.